

EDUCATION

University of Adelaide

Bachelor of Computer Science, Major in Cybersecurity

Adelaide, SA

Feb 2022 - Dec 2025

- GPA: **6.4/7.0** – Achieved 11 High Distinctions across core subjects
- Top Results: Databases & Ethical Data (96), Web & Database Computing (95), Algorithm & Data Structure Analysis (93), Software Engineering & Project (92)

EXPERIENCE

Profilarr ([GitHub](#), [Website](#))

April 2023 - Present

Creator & Maintainer

- GitOps-inspired tool for declarative configuration management—syncs version-controlled configurations to self-hosting tools, resolves merge conflicts, and preserves customizations. Built with **SvelteKit** and **Deno** (migrated from React/Flask).
- Scaled to **1,000,000+ downloads** and **30,000+ active users**; built community with **1,700+ GitHub stars**, **200+ forks**, and **1,200+ Discord members**.
- Designed **PCD specification** (standardized format for shareable configuration databases) and **OSQL**, an append-only event-sourcing approach that stores changes as replayable operations—enabling conflict detection and seamless upstream updates. Built **Rosettarr** translation layer for YAML-to-OSQL migration.
- Documentation platform statically generated from database content—tracks configuration changes, recommends profiles, hosts dev logs and wiki. Designed to be forkable for other databases. Evolved through MkDocs → Next.js → **Svelte**.
- Discord bot (**TypeScript**, Discord.js) for polling, search, automated changelogs, and moderation.

PROJECTS

OMesh ([GitHub](#))

Python, Docker, WebSockets

Secure Messaging Platform

- Built end-to-end encrypted messaging platform using low-level socket programming and custom protocol design.
- Implemented RSA/AES-GCM cryptographic pipeline for secure key exchange and message encryption.
- Deployed multi-server architecture with Docker Compose; real-time communication via WebSockets.

Synapsis ([GitHub](#))

React, Express, LLMs, FFmpeg

AI Research Paper Video Platform

- Built full-stack platform using Grobid for paper parsing, LLMs for summarization/image generation, and FFmpeg for video synthesis.
- Pitched at **Tech eChallenge** startup competition; ranked **Top 5** out of 50+ teams.

Rogue ([GitHub](#))

C++, SFML

Procedural Roguelike Game

- Built turn-based roguelike with procedural map generation and entity component system architecture.
- Implemented design patterns (Factory, Observer, State) for extensible game object behavior.

Episode Splitter ([GitHub](#))

Python, scipy, FFmpeg

Audio Cross-Correlation Tool

- Built tool using audio cross-correlation to automatically detect episode boundaries in multi-episode files with millisecond precision—processing 146 episodes in 20 minutes vs weeks manually.

SKILLS

Languages: TypeScript, Python, C++, SQL, HTML/CSS

Technologies: SvelteKit, Deno, Docker, Git, Linux, WebSockets, REST APIs

Practices: GitOps, Event Sourcing, API Design, Open Source Development

Interests: Self-Hosting & Homelabs, Basketball, Technical Writing